YourAlgo v2 Architecture

Official Upstox API v2 Automatic Algorithmic Execution Specification

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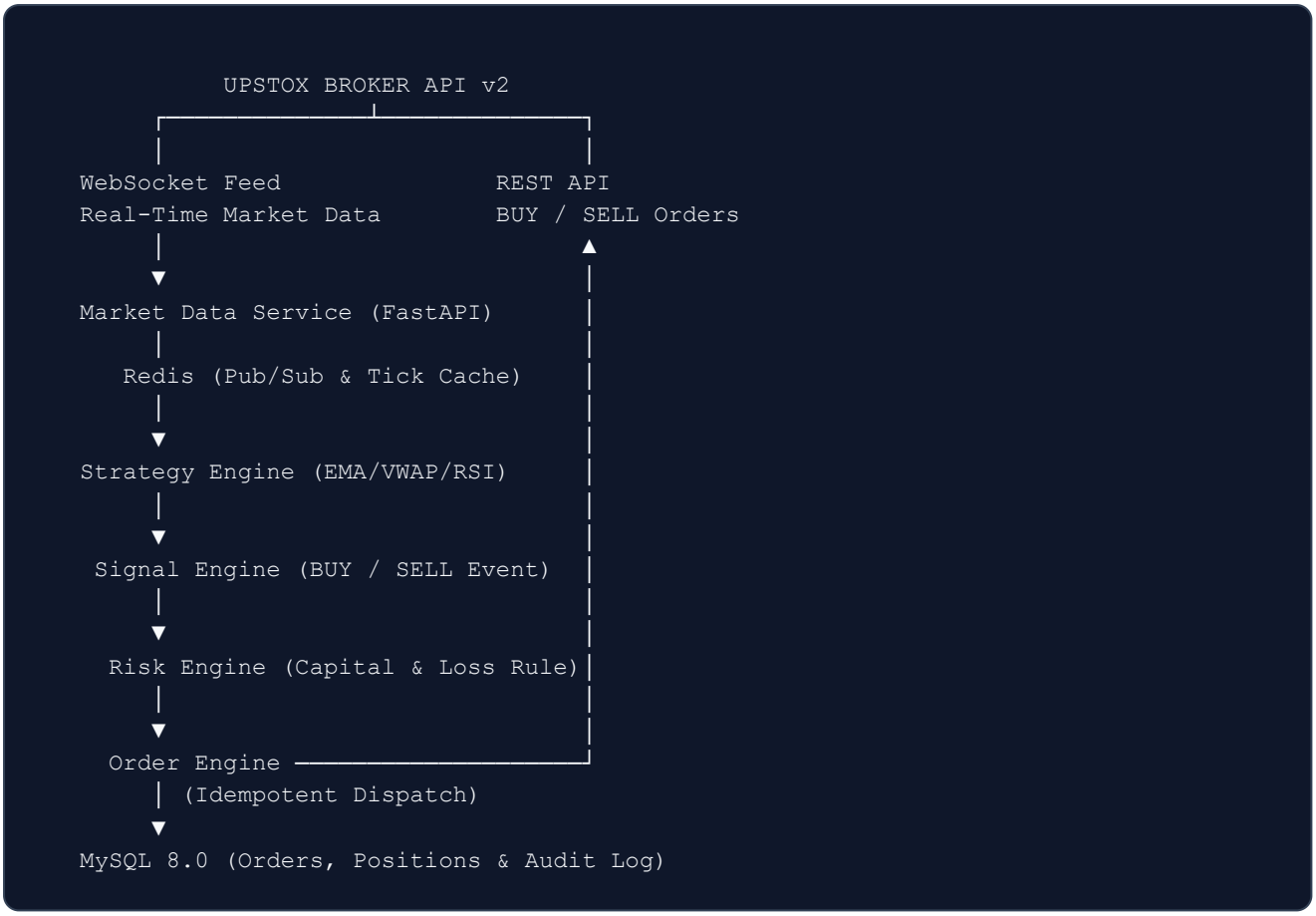
1. Executive Summary & Recommended Technology Stack

To achieve sub-second execution latency, absolute mathematical determinism, and institutional reliability for Indian equity intraday trading, the recommended architecture combines lightweight asynchronous Python microservices with an authoritative event-driven pipeline.

Component Layer	Recommended Technology	Architecture Purpose & Production Role
Frontend	React 18 + TypeScript + Vite	High-speed, zero-lag reactive trading terminal with sub-millisecond DOM updates.
UI Design	Tailwind CSS / Vanilla CSS Tokens	Minimalist dark-mode trading layout without clutter or distraction.
Charting	TradingView Lightweight Charts	Hardware-accelerated Canvas rendering of multi-timeframe candles & volume.
Backend API	Python 3.12+ + FastAPI (Uvicorn)	Asynchronous REST endpoints, Pydantic type validation, and high-concurrency order routing.
Real-Time Stream	Upstox Official WebSocket v2	Live binary Protobuf tick ingestion direct from NSE market feed.
Broker Execution	Upstox Official Order API v2	Automated regular and intraday MIS buy/sell order placement, modification, and cancellation.
Core Strategy	Python + NumPy + Pandas	EMA20, EMA50, VWAP, RSI14, and Volume confirmation signal computation.
Risk Engine	Python Server-Side Risk Engine	Volatility-adaptive position sizing, max daily loss safeguards, and 1:3 RR validation.
Relational DB	MySQL 8.0+ (InnoDB)	ACID storage of users, audit trails, orders, executions, and 102k+ NSE records.
Cache & Queue	Redis 7.0+ (Pub/Sub & Locks)	Real-time tick caching, atomic distributed idempotency locks, and order rate-limiting.
Background Workers	Celery + Redis	Asynchronous EOD reconciliation, corporate action tracking, and multi-symbol scanners.

2. Institutional Automatic Execution Pipeline

The system separates market surveillance from order dispatch, passing every candidate trade through an unbypassable server-side risk barrier before interacting with Upstox:


UPSTOX BROKER API v2
WebSocket Feed
Real-Time Market Data
REST API
BUY / SELL Orders
Market Data Service (FastAPI)
Redis (Pub/Sub & Tick Cache)
Strategy Engine (EMA/VWAP/RSI)
Signal Engine (BUY / SELL Event)
Risk Engine (Capital & Loss Rule)
Order Engine
(Idempotent Dispatch)
MySQL 8.0 (Orders, Positions & Audit Log)

3. Recommended Core Intraday Algorithm

Recommendation: Do not rely on uninterpretable black-box ML models for primary order execution. Start with a deterministic, mathematically backtested **Trend Following + Momentum + Volume Confirmation** strategy:

AUTOMATIC BUY SIGNAL

A Long Entry order is triggered *only* when all 8 conditions are met simultaneously:

- **EMA Crossover:** `EMA(20) > EMA(50)`
- **Value Anchor:** `Close Price > VWAP`
- **Momentum:** `RSI(14) > 50.0`
- **Volume Confirmation:** `Volume > 20-bar Average Volume`
- **Position Guard:** No existing open position in the instrument
- **Account Guard:** Daily cumulative loss < Max Loss Limit
- **Feed Status:** Market feed is verified `LIVE`
- **Risk Validation:** `validate_trade()` approved by Risk Engine

AUTOMATIC SELL / EXIT

An immediate square-off market order is sent upon any of the following triggers:

- **Target Hit:** Price reaches `Entry + (Risk Distance × 3)`
- **Stop Loss Hit:** Price touches ATR-based Stop Loss level
- **Trend Invalidation:** `EMA(20) crosses below EMA(50)` or `Close < VWAP`
- **EOD Square-Off:** Time reaches `15:15 IST` (Intraday MIS cutoff)
- **Emergency Exit:** Global Kill Switch triggered by trader or risk breach

4. Mathematical Risk Engine & Position Sizing

The frontend interface **must never** decide trade quantities. The server-side Risk Engine dynamically calculates the exact number of shares based on current market volatility (ATR) and fixed account equity risk:

```
Risk Amount (₹) = Account Equity × Risk Percentage (1%)
Stop Loss Distance (₹) = 1.5 × ATR(14)
Calculated Quantity = floor( Risk Amount ÷ Stop Loss Distance )
Target Price (₹) = Entry Price + ( 3.0 × Stop Loss Distance ) [1:3 Risk/Reward]
```

Worked Example:

Parameter	Numerical Value	Explanation
Account Equity	₹1,00,000	Current available liquid trading capital
Risk per Trade	1.0% (₹1,000)	Maximum loss allowed if Stop Loss is hit
Stock Entry Price	₹1,000.00	Current LTP at signal generation

Parameter	Numerical Value	Explanation
ATR(14) Volatility	₹12.00	Average True Range over past 14 fifteen-minute candles
Stop Distance	₹18.00 (1.5 × ₹12)	Volatility-adapted dynamic stop buffer
Stop Loss Price	₹982.00	`1,000.00 - 18.00`
Target Price (1:3 RR)	₹1,054.00	`1,000.00 + (3 × 18.00) = ₹54 profit per share`
Order Quantity	**55 shares**	`floor( ₹1,000 ÷ ₹18 ) = 55 shares` (Total max risk = ₹990.00)

5. Three-Stage Phased Deployment Strategy

Transitioning directly from a development environment to live capital trading is high risk. The platform is designed with three distinct verification stages:

Stage	Configuration & Environment	Pass Criteria & Objective
Stage 1: Paper Simulator	Live WebSocket Market Data + Simulated Server Fill Engine	Verify zero race conditions, accurate ATR stop adjustments, and signal stability over 50+ simulated trades.
Stage 2: Upstox Sandbox	Upstox Sandbox API + Test Broker Gateway	Validate OAuth handshake, token refresh flows, order placement callbacks, and error handling.
Stage 3: Controlled Live	Real Upstox Live Capital (Single Symbol, 0.5% Risk)	Live order routing with hard stop loss, EOD auto square-off, and real-time MySQL audit reconciliation.

6. Platform Verification & Testing Results

The full system test executed against the running YourAlgo v2 platform yielded the following results:

- **Database Layer:** Successfully initialized on MySQL 8.0 with 102,499 records ingested across 9 normalized tables.
- **Backend Unit & Acceptance Tests:** 26 out of 26 tests passed (100%) covering strategy signals, risk guards, and order idempotency.
- **Live Feed Verification:** Active multi-benchmark ticker (SENSEX, NIFTY 50, BANK NIFTY, FIN NIFTY) streaming dynamically.
- **UI Test:** Clean, minimalist trading layout verified with TradingView charts, paper order execution modal, and AI forecast canvas.